

Human-Computer Interaction

Autumn 2026

Assignment 1 – Empirical Analysis of Fitts' Law

Due: Wednesday, September 16, 2026, by 11:59 AM (for Part I)

Wednesday, September 23, 2026, by 9:00 AM (for Part II)

OBJECTIVE: In this assignment, you will validate Fitts' Law by generating your own pointing task dataset, performing statistical regression, and deriving your personal predictive motor-control model.

PART I. Classical Fitts' Law Tapping Experiment

Step 1. Data Collection (record your screen during the experiment, upload to Youtube or Drive, and make the video publicly accessible)

Navigate to the CS Field Guide Fitts' Law experiment:

<https://www.csfieldguide.org.nz/en/interactives/fitts-law/>.

Complete a full experimental session to ensure a statistically significant sample size (100 trials by default). Export your results as a CSV file. Ensure your dataset includes Distance (A), Width (W), Index of Difficulty (ID), and Average Time (MT).

Step 2. Parameter Calculation & Analysis

Using a statistical tool of your choice (Python, R, Excel, etc.), perform a linear regression on your dataset to find the coefficients a (intercept) and b (slope) for the standard Shannon formulation:

$$MT = a + b \log_2 \left(\frac{A}{W} + 1 \right)$$

Provide your formula: Write out the complete Fitts' Law equation using your empirically derived values for a and b .

Explain the parameters: In a brief paragraph, explain what your specific a and b values represent in the context of human motor control and reaction time. What real-world factors might cause these numbers to be higher or lower for different users?

Step 3. Data Visualization

Generate a scatter plot of your experimental results.

Map the **Index of Difficulty (bits)** to the x-axis and **Movement Time (ms)** to the y-axis.

Overlay your linear regression trendline (the line of best fit) onto the scatter plot.

Requirement: Your chart must be formatted to an academic standard. Include clear axis labels with units, a descriptive title, and display your R^2 correlation score on the plot to indicate the strength of your model.

What to Submit? Please compile your work into a single PDF file that documents each step of your process. Your PDF must include the following:

- **Experiment Video Link:** A public link to your screen recording (YouTube/Drive).
- **Trial Data Table:** A clear, embedded table showing your collected dataset (Distance, Width, ID, and MT). Do not upload a separate CSV.
- **Fitts' Law Formula & Analysis:** Your calculated equation and a short explanatory paragraph analyzing your specific a and b parameters.
- **Scatter Plot:** Your formatted visualization containing the trendline and R^2 score.

PART II. AI-Assisted Fitts' Law Experiment & Empirical Analysis

You will transition from consuming HCI theory to creating interactive experimental tools. You can use Generative AI (Gemini, Claude, or ChatGPT) as a rapid-prototyping partner to build a custom, web-based Fitts' Law testing application. You will deploy this app, use it to gather human performance data, and perform a simple analysis just like previous tasks.

Step 1. Ideation and AI-Assisted Prototyping

Instead of writing HTML/JS from scratch, your first task is to practice “prompt engineering” to generate a working experiment. Before writing any code, you must design the concept for your Fitts' Law variant. As graduate students, you should ground your experiment in a real-world design problem. **So, define the scenario first.** Who is the user and what is their context? e.g., A driver adjusting the A/C temperature on a touchscreen while keeping their eyes on the road.

Prompt an LLM: Use an AI tool (Gemini, Claude, ChatGPT) to generate your application based on your concept.

Requirements: Your generated single-file HTML/JS application must include: An interactive canvas reflecting your chosen Scenario and Innovation. Logic that systematically varies Target Distance and Target Width. An accurate timer recording Movement Time. A feature to export the trial data (A, W, and MT) as a CSV file.

Step 2. Deployment

Host your experiment publicly using GitHub.

Step 3. Empirical Analysis

Next, act as the participant to collect your data. Record your screen during the trials, embed this video directly into your GitHub page, and perform your analysis by generating an academic-quality scatter plot of the results.

Step 4. Report

On your Github page, briefly describe your Scenario, Innovation, and Application. Why did you choose this design, and what real-world HCI problem does it attempt to solve?

Your Custom Formula: Write out the complete Fitts' Law equation using your empirically derived values for a and b .

What to Submitt? Please submit your live GitHub Pages URL (e.g., [https://\[username\].github.io/\[repo-name\]/](https://[username].github.io/[repo-name]/)). You can do this by copying and pasting the link into the text submission box on eclass.